

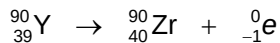
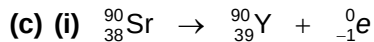
8 (a) (i) Spontaneous decay means the decay process is not affected by any external factors such as chemical composition and physical environment (temperature, pressure, electric fields, magnetic fields, luminosity, etc.)

(ii) Random decay means that it is impossible to predict exactly which nucleus and when a particular nucleus will disintegrate, although statistically, it is possible to predict the probability of decay per unit time if its decay characteristics are known.

(b) (i) Activity, $A = -\frac{\Delta N}{\Delta t}$.

(ii) Probability of decay in time $\Delta t = \frac{\left(-\frac{\Delta N}{\Delta t}\right)}{N}$.

(iii) The decay constant $\lambda = \text{probability of decay in time } \Delta t = \frac{\left(-\frac{\Delta N}{\Delta t}\right)}{N}$.



(ii) Decay constant, $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{65 \times 60 \times 60} = 2.96 \times 10^{-6} \text{ s}^{-1}$.

(d) (i) Using $N = N_0 e^{-\lambda t}$, fraction of strontium-90 remaining, $\frac{N}{N_0} = e^{-\frac{\ln 2}{28} \times 50} = 0.29$.

- (ii) The half-life of Strontium-90 is 28 years while half-life of yttrium is 65 hours. As such, when a Strontium-90 nuclei decays to produce a yttrium isotope, the yttrium will then decay in a comparative short time after its formation to produce a zirconium nuclei as its half-life is only 65 hours. Thus, the number of zirconium nuclei is almost equal to the number of strontium nuclei that have decayed.
- (iii) β - particles are fast moving electrons and the β - particles are emitted collide with the atoms of the lead container. These collisions result in the β - particles undergoing deceleration, either being slowed down by collision or by deflection. The electrons will then release energy in the form of the continuous spectrum of the X-ray produced, also called Bremsstrahlung radiation.